

Smart Home Security System with Automatic Phone Calling using Arduino & IoT

OBJECTIVE

To design and implement a smart home security system that detects unauthorized entry and automatically alerts the user through phone calls using Arduino and IoT technology.

ABSTRACT

This project presents a smart home security system using Arduino integrated with IoT technology. The system detects intrusions using sensors such as PIR (motion sensor) or door sensors. When an unauthorized event is detected, the system automatically triggers an alert mechanism such as making a phone call to the user using a Bluetooth module.

The system enhances home safety by providing real-time alerts and remote monitoring. It is cost-effective, easy to install, and can be expanded with additional smart features

INTRODUCTION

With increasing security concerns, smart home systems are becoming essential. Traditional security systems lack real-time alerts and remote access. This project overcomes these limitations by integrating Arduino with IoT and communication modules.

The system continuously monitors the home environment and immediately informs the user in case of suspicious activity, ensuring quick response and improved safety.

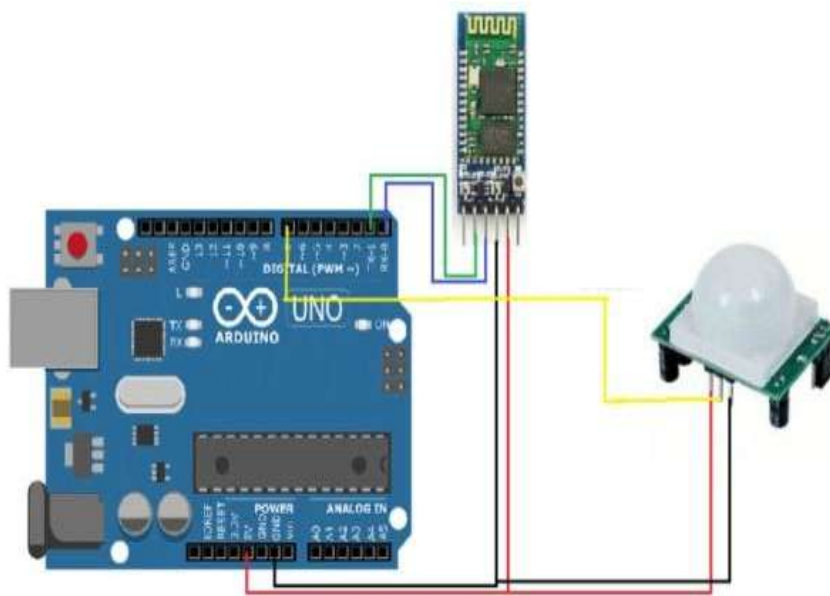
HARDWARE REQUIREMENT/DESCRIPTION

- Arduino Uno: A microcontroller board that controls the system by processing sensor input and sending signals.
- PIR Motion Sensor: Detects motion by sensing changes in infrared radiation from nearby objects.
- Bluetooth Module (HC-05): Enables wireless communication between Arduino and a mobile device.

CIRCUIT/COMPONENT SPECIFICATIONS

Operating Voltage	5 V
PIR Sensor Range	3 to 7 meters
Bluetooth Module	HC 05

CIRCUIT DIAGRAM



DESIGN FORMULA

- The system operates at **5V DC** with total power consumption $\approx 0.18 \text{ W}$.
- The PIR sensor gives a **digital output (0–5V)** to the Arduino when motion is detected.
- Motion is detected within a range of **3–7 meters** and $\sim 120^\circ$ angle.
- The Arduino sends a signal via **Bluetooth (HC-05)** to the mobile phone.
- The mobile app triggers an **automatic call/SMS alert** for security notification. Maximum supply voltage should not exceed 15V
- Humidity should not exceed 85% relative humidity.
- Timing tolerance should not exceed +10 sec for 1 min.

APPROACH/METHODOLOGY

The system continuously monitors motion using a PIR sensor. When motion is detected:

1. Sensor sends signal to Arduino
2. Arduino processes the input
3. Bluetooth module makes an automatic phone call or sends SMS
4. User receives alert and takes action

CONCLUSIONS

Thus, the smart home security system is successfully implemented. It provides real-time alerts through automatic phone calls and ensures enhanced home safety. The system is reliable, cost-effective, and scalable for future improvements.

REFERENCES

- <https://www.electronicsforu.com/electronics-projects/smart-home-security-system>
- <https://appinventor.mit.edu/>

APPENDIX

PIR SENSOR



Detects infrared radiation emitted by human bodies and senses motion.

BLUETOOTH MODULE



Used for communication. It makes calls using a SIM card.

ARDUINO UNO



Microcontroller board used to control the entire system.

PHOTO OF EXPERIMENTAL IN WORKING CONDITION

